Department of Computer Science and Engineering

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch: B.E., CSE, V Semester

Course Code & Title: CCS375 Web Technologies

Name of the Faculty member: Dr. S. Poornam

Innovative Practice Description

Unit / Topic: Unit I/ Cascading Style Sheets (CSS3)

Course Outcome: CO1

Topic Learning Outcome: TLO5

Activity Chosen: Roleplay Activity

Date & Time: 06.08.2025 & 09:10 A.M

Justification

This roleplay activity helps students visualize the concept of background, content, and layering in CSS3 through a physical classroom demonstration. By acting as elements (background, text, and image) and rearranging themselves, students understand how CSS3 controls element stacking using z-index and positioning. The activity fosters active learning, peer collaboration, and experiential understanding, bridging abstract CSS3 concepts with real-world web design scenarios.

Time Allotted for the Activity: 20 Minutes

Details of the Implementation

- Three students were chosen to represent different webpage layers:
  - Student 1&2 = Background
  - Student 3 = Text
  - Student 4 = Image
- Initially, students stood in a straight line representing a plain stacking order.
- The instructor then rearranged their positions (background behind, text in the middle, image in front) to simulate the layering of elements.
- Students were asked to imagine how a webpage would look if these positions changed, showing how CSS3's z-index property controls visibility and stacking order.

- Additional variations included:
    - Placing text behind the image (hidden effect).
    - Bringing the background forward (illogical in real design).
- Students actively participated by swapping roles and experimenting with different "layer orders."

CO – PO / PSO Mapping

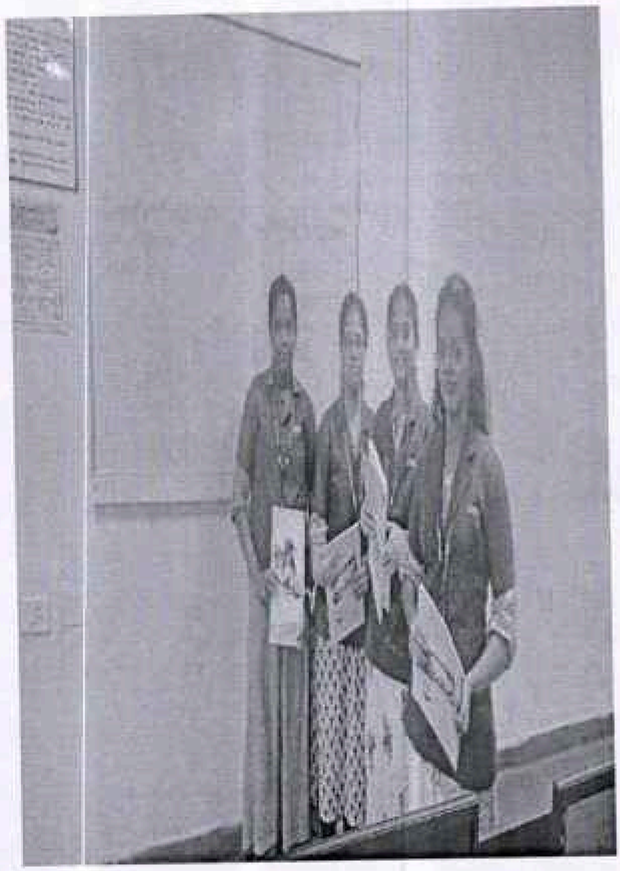
CO	PO1	PO2	PO3	PO4	PSO1
CO1	2	2	2	1	1

(1 – Low, 2 – Moderate, 3 – High)

Innovative practice	PO1	PO2	PO3	PO4	PSO1
	2	2	2	1	1
Justification for correlation	Students gained a strong understanding of the fundamental principles of Cascading Style Sheets, including the cascade, specificity, and the box model.	Students analysed real-world webpage layouts and design problems to identify suitable CSS properties and techniques.	Students designed and developed responsive and visually appealing webpage interfaces using CSS layouts, selectors, and styling rules.	Students explored complex styling challenges such as responsive grids, animations, and cross-browser adjustments to refine their solutions.	Students applied their CSS knowledge in practical exercises like the Roleplay Activity, demonstrating their ability to create professional and engaging web designs.

PO/PSO mapped: PO1, PO2, PO3, PO4, PSO1

Images / Screenshot of the practice:

Reflective Critique:

- Students reported that the roleplay activity made the concept of CSS3 layering much easier to understand compared to other illustrative methods.
- Students appreciated the hands-on, visual nature of the activity, which helped them grasp the abstract idea of element stacking and z-index manipulation more intuitively.
- Students expressed that the physical demonstration of layering improved their comprehension of how backgrounds, text, and images interact on a webpage.

Benefits of the practice:

- This activity successfully enhanced students' understanding of CSS3 layering principles by linking theoretical knowledge to a tangible experience.
- This activity fostered better conceptual clarity about the stacking context and z-index property.

➢ Additionally, student's coding skills improved as they could relate the classroom demonstration to actual CSS code more effectively.

➢ The interactive nature of swapping roles encouraged active learning and peer collaboration, deepening engagement and retention of the concepts.

Challenges faced in implementation:

➢ A notable challenge was the limited participation, with only a few students actively involved at a time.

➢ This restricted the full engagement of the entire class, potentially reducing the collective learning impact.

➢ Finding ways to involve more students simultaneously or through parallel activities could enhance inclusivity and maximize benefits.

References

- https://css-tricks.com/what-you-may-not-know-about-z-index/
- https://www.w3schools.com/css/css_zindex.asp
- https://en.wikipedia.org/wiki/Role-playing
- https://www.youtube.com/watch?v=UAAPQ6wxxis
- https://serc.carleton.edu/introgeo/roleplaying/whatis.html

S. [signature] 17/10/25

Signature of Faculty Member

[signature]

HOD